Checking domain on Non-linear equations

$$\frac{\sqrt{x^2-4}}{7x-7}$$

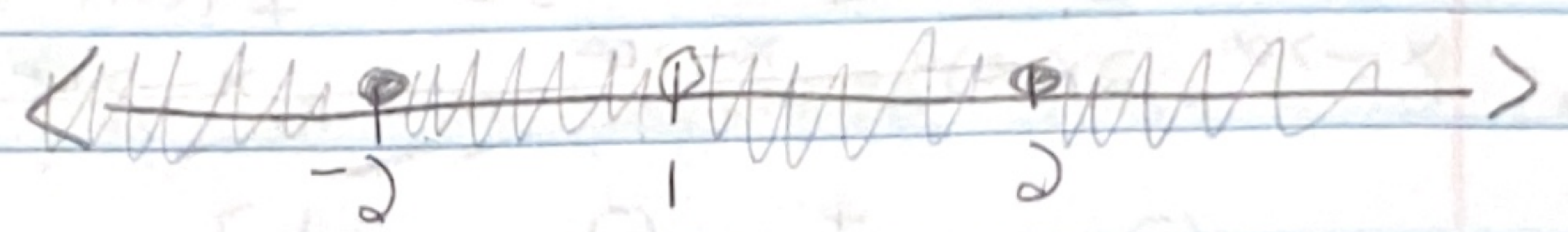
We know we cannot have a negitive in the square root and we also know the denominater is not a zero we have two operation here the square root and the denominator

[take the domain of the one with the most restrictions]

$$x^2 - 4 \geq 0$$
$$+4 \quad +4$$

$$\sqrt{x^2} \geq \sqrt{4}$$

$$x \geq \pm 2$$



$(-\infty, -2] \cup [-2, 1) \cup (1, 2] \cup [2, \infty)$

$$7x - 7 \neq 0$$
$$+7 \quad +7$$

$$\frac{7x}{7} \neq \frac{7}{7}$$

$$x \neq 1$$

- after getting the domain we will pick number from in between them to plug into the equation to see which one violates our rules so we can cross it out.

✓ $(-\infty, -2] = -3$ $\quad \frac{\sqrt{(-3)^2-4}}{7(-3)-7} = \frac{\sqrt{5}}{-28}$ (No negitive in sq root) (No zero in denominator)

X $[-2, 1) = 0$ $\quad \frac{\sqrt{(0)^2-4}}{7(0)-7} = \frac{\sqrt{-4}}{-7}$ (Negitive is in sq root) (No zero in denominator)

X $(1, 2] = 1.5$ $\quad \frac{\sqrt{(1.5)^2-4}}{7(1.5)-7} = \frac{\sqrt{-1.75}}{3.5}$ (Negitive is in sq root!!) (No zero in denominator)

✓ $[2, \infty) = 3$ $\quad \frac{\sqrt{(3)^2-4}}{7(3)-7} = \frac{\sqrt{5}}{14}$ (No negitive in sq root) (No zero in denominator)

Domain = $(-\infty, -2] \cup [2, \infty)$

You must do this with non-linears equation but it is also recommend that you do this with all domain that you find limit of weirdness

① $\lim_{x\to\infty} \left(\frac{9}{2x^5} + 3x^{-4} - 13\right)$ distribute the limit

$\lim_{x\to\infty} \frac{9}{2x^2} + \lim_{x\to\infty} 3x^{-4} - \lim_{x\to\infty} -13$

$\lim_{x\to\infty} \frac{9}{2(\infty)^2} + \lim_{x\to\infty} 3(\infty)^{-4} + \lim_{x\to\infty} -13$

$0 + 0 \quad -13 \quad = \boxed{-13}$

② $\lim_{x\to\infty} \left(e^{-x} + \ln\left(e + \frac{1}{x}\right)\right)$

$\lim_{x\to\infty} e^{-x} + \lim_{x\to\infty} \ln\left(e + \frac{1}{x}\right)$

$\lim_{x\to\infty} e^{-(\infty)} + \ln\left(\lim_{x\to\infty} e + \lim_{x\to\infty} \frac{1}{\infty}\right)$

$0 + \ln(e + 0)$

$\boxed{\ln e = 1}$

③ $\lim_{x\to -\infty} (e^x \sin)$ - You can factor out the e^x and the sin can only go from [-1, 1]

$-1 \le \sin < 1 \quad \cdot e^x$

$-e^x \le e^x \sin \le e^x$

$\lim_{x\to -\infty} -e^x \le \lim_{x\to -\infty} e^x \sin \le \lim_{x\to\infty} e^x$

$0 \le \lim_{x\to -\infty} e^x \sin \le 0 \quad \longrightarrow$

$-1 \le \sin \le 1$ $\cdot e^x$

$-e^x \le e^x \sin \le e^x$

$\lim_{x \to -\infty} -e^x \le \lim_{x \to -\infty} e^x \sin \le \lim_{x \to -\infty} e^x$

$0 \le \lim_{x \to -\infty} e^x \sin \le 0$

the $\lim_{x \to -\infty} e^x \sin$ is in between zero and zero

So the answer will just be zer

$\lim_{x \to -\infty} e^x \sin = 0$

$\lim_{x \to \infty} \frac{\cos 3x}{5x^2+1}$

- a fraction is a whole term
- cos can only go from [-1, 1]

$-1 \le \cos \le 1$ $\left(\frac{3x}{5x^2+1}\right)$

You cannot apply a ∞ limit to a sin or cos because they only go from [-1, 1]

$\frac{-3x}{5x^2+1} \le \frac{\cos 3x}{5x^2+1} \le \frac{3x}{5x^2+1}$

$\lim_{x \to \infty} \frac{-3x}{5x^2+1} \le \lim_{x \to \infty} \frac{\cos 3x}{5x^2+1} \le \lim_{x \to \infty} \frac{3x}{5x^2+1}$

$0 \le \lim_{x \to \infty} \frac{\cos 3x}{5x^2+1} \le 0$

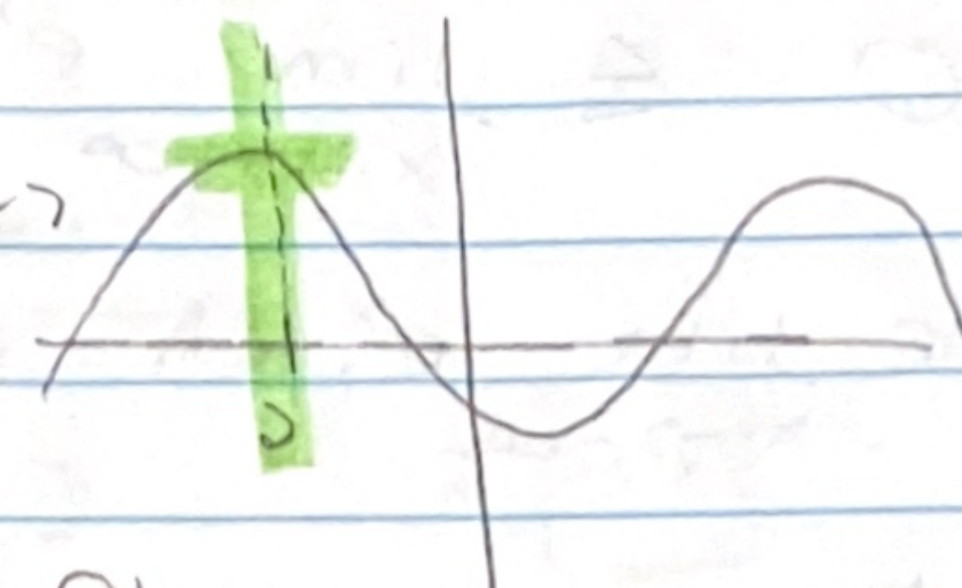
$\lim_{x \to \infty} \frac{\cos 3x}{5x^2+1} = 0$

$f'(x) = DNE$

- Sharp turn
- Not continous
  - hole
  - VA
  - Jump
- vertical asymptote

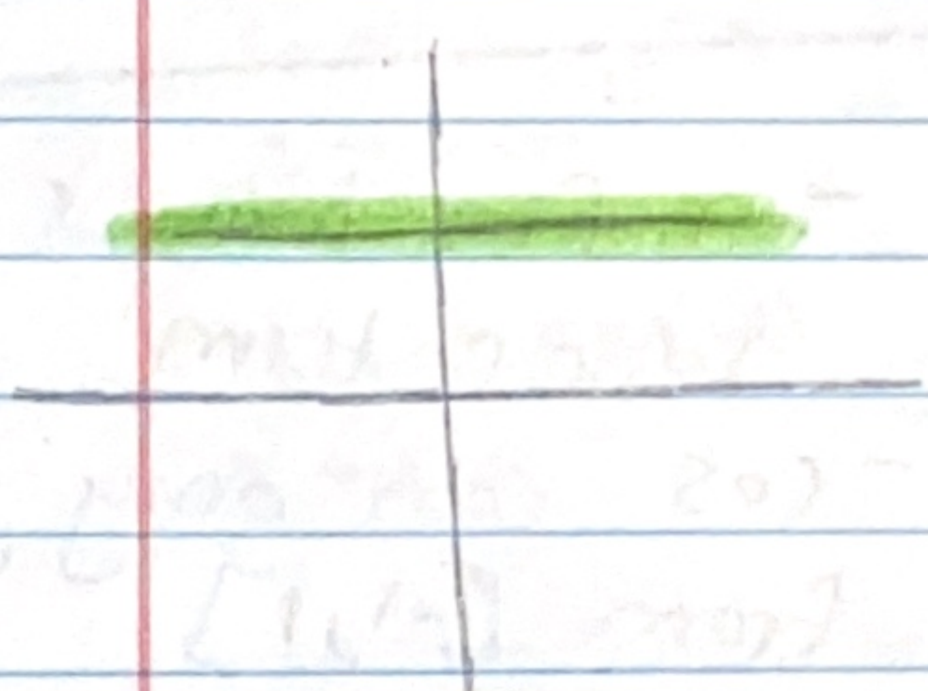
Smooth turn $f'(x) = 0$

- $f'(0) = 0$ Smooth turn →
(horizantal tangetline)



$f'(2) = 0$

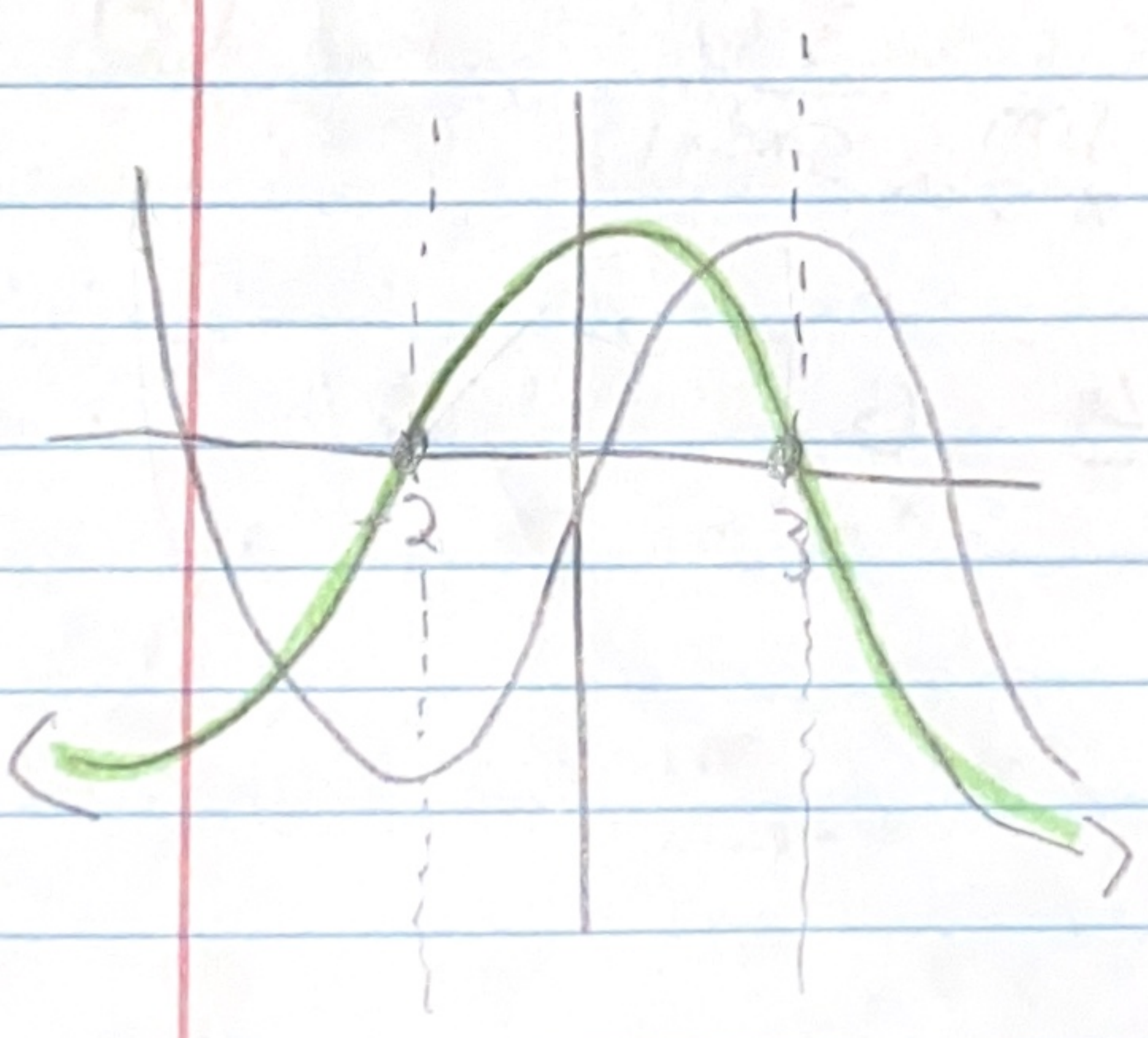
$f'(x) = 0$ if the slope of the line is just strignt example:



there is no rise or run
so slope is 0

read graph from left to right

if $f'(x)$ is decreasing then line below x-axis
if $f'(x)$ is increasing then line above x-axis



began from the smooth curve